Xi Chen

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Profile

Statistics PhD candidate with experience building machine-learning algorithms and reproducible training and evaluation workflows for graph-structured, time-series, and biomedical data. Skilled in data preprocessing, PyTorch model development, numerical optimization, benchmarking, and statistical evaluation.

Education

University of California, Santa Cruz	PhD, Statistics; candidacy: 07/2026	2023–Expected 06/2028
University of California, Davis	MS, Biostatistics	2021–2023
Sun Yat-Sen University	BS, Applied Mathematics	2014–2018

Technical Skills

Programming & Data: Python (NumPy, pandas, PyTorch, scikit-learn), CUDA, R (Rcpp), SQL, MATLAB
Machine Learning: CNNs, LSTMs/RNNs, Transformers, random forest, gradient-boosted trees, k-means, time-series analysis, multivariate analysis, changepoint detection, Bayesian statistics
Algorithms & Tools: ADMM, MCMC (MH/Gibbs sampling), Git, SSH, Shell/Linux, Jupyter

Relevant Experience

PhD Candidate & Graduate Researcher Sep 2023–Present
University of California, Santa Cruz

Latent-Space Graph-Fused LASSO [*arXiv preprint*]

- Co-developed unsupervised clustering that combines graph links and node features. Implemented a PyTorch neural decoder, graph-fused ADMM optimization, and Langevin sampling to learn clusters.
- Benchmarked against classical and neural graph-clustering methods: reached 97.0% normalized mutual information versus 83.1% for the strongest listed baseline on a 144-node simulated grid. Combined county borders with 168 monthly temperature observations to identify 10 climate regions among 58 California counties.

GraphClustAR

- Co-developing interpretable clustering of graph-linked time series through node-specific autoregressive models; implementing graph-fused ADMM updates, BIC model selection, and an R package.

Multivariate Changepoint Detection

- Co-developed tests for a shared change in the means of correlated time series, allowing each variable to change by a different amount or direction. Implemented weighted vector U-statistics to detect changepoints near the beginning or end of a time series, and VAR pre-whitening to account for serial dependence.
- Built Monte Carlo evaluations with up to 300,000 simulations per setting to assess false-positive control and detection power; analyzed cardio-respiratory data from a CO_2 inhalation experiment. Rcpp made the adjusted simulation workload about 4× faster than an R reference in a 1,000-run benchmark.

Research Assistant Sep 2019–July 2021
Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences

- **Blood-Pressure Estimation:** Trained supervised deep learning models to estimate blood pressure from ECG/PPG waveforms for potential wearable monitoring; filtered signals, segmented and aligned waves, and removed artifacts before model training.
- **Arrhythmia Detection:** Developed and evaluated a PPG-based CNN for six-rhythm classification using ECG-informed labels. The co-authored *JAHA* study reported 85.0% accuracy and 0.978 micro-AUROC on 23,384 held-out waveforms.

Selected Publications & Preprints

- Kei, Y. L., Padilla, O. H. M., Killick, R., Wilson, J., **Chen, X.**, and Lund, R. (2025). Decoder-only clustering in attributed graphs. arXiv:2511.04859.
- Liu, Z., Zhou, B., Jiang, Z., **Chen, X.**, et al. (2022). Multiclass arrhythmia detection and classification from photoplethysmography signals using a deep convolutional neural network. *Journal of the American Heart Association*, 11(7).